

Sound Before Light: The Baryon Acoustic Ruler as a Frozen Sound Horizon Computed from First Principles

TZPID Gold Spine Series II, Paper XI of XX

Daniel Alexander Trawin
ORCID: 0009-0001-4630-3715
DOI: 10.5281/zenodo.20723171
danielalexandertrawin@gmail.com

June 2026

Abstract

The second series of the TZPID spine confronts the breathing-enclosure cosmology with real data, and opens—as the first series did—on its most defensible quantitative node. The baryon acoustic oscillation (BAO) standard ruler that anchors modern cosmological distance measurement is a frozen sound wave: a pressure oscillation in the primordial photon–baryon plasma whose comoving scale is set before recombination releases the cosmic microwave background. We compute the comoving sound horizon directly from the acoustic integral $r_s = \int_z^\infty c_s(z')/H(z') dz'$, with $c_s = c/\sqrt{3(1+R_b)}$ and neutrino-corrected radiation $\Omega_r = \Omega_\gamma(1 + 0.2271 N_{\text{eff}})$, $N_{\text{eff}} = 3.046$. The result reproduces the Planck 2018 value to 0.1% at last scattering (144.2 vs 144.4 Mpc) and 0.2% at the drag epoch (146.9 vs 147.1 Mpc). The disciplined consequence is the sound-before-light premise: acoustic physics lays down the metric ruler before the oldest freely streaming light is released. We make the slogan precise— $c_s < c$ throughout, no superluminal propagation, and a hot dense plasma is required, so sound-before-light confirms rather than removes the hot early universe. The first node of the registry standing wave $\delta_b \sim A j_0(kr_s)$ sits at $kr_s = \pi$, the fundamental enclosure mode of Paper I, anchoring the result to ID7257, ID7259, ID7733 and supplying the acoustic clock for $H_0 = \dot{R}/R$ (ID10867).

1 Introduction: the ruler is older than the light

A familiar reading of cosmic origins places light first. The early universe says otherwise: the first imprinted standard is acoustic, not luminous. Before recombination the universe was a single tightly coupled photon–baryon fluid; photons were scattered so frequently that the plasma behaved as one elastic medium. Pressure waves propagated outward from every overdensity until, at recombination, the plasma went neutral, the photons decoupled, and the wave pattern froze into the matter distribution. That frozen pattern is the BAO ruler, observed as the $\sim 100 h^{-1}$ Mpc feature in galaxy clustering and as the CMB acoustic peaks. The first series (Papers I–VIII) read this as the acoustic interior of a bounded enclosure; the present paper makes the underlying quantitative claim testable at the percent level.

2 The acoustic premise, made precise

The slogan admits a sloppy reading (sound before photons—false, photons are present throughout) and a precise one:

$$\boxed{\text{Acoustic pressure modes set the metric ruler before the oldest freely travelling light was released.}} \quad (1)$$

The relevant light is the CMB. The ruler it carries was imprinted by sound during the preceding plasma epoch: the fundamental radial mode of the outermost S^3 , frozen at decoupling.

3 The sound speed and the acoustic horizon

The photon–baryon sound speed is

$$c_s(a) = \frac{c}{\sqrt{3(1+R_b(a))}}, \quad R_b(a) = \frac{3\Omega_b}{4\Omega_\gamma} a, \quad (2)$$

bounded above by $c/\sqrt{3} \approx 0.577c$: strictly subluminal. The comoving sound horizon at redshift z_* is

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)} dz, \quad H(z) = H_0 \sqrt{\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_\Lambda}, \quad (3)$$

with radiation including neutrinos, $\Omega_r = \Omega_\gamma(1 + 0.2271 N_{\text{eff}})$, $N_{\text{eff}} = 3.046$. Omitting the neutrino term inflates the horizon by $\sim 10\%$.

4 Result: the horizon to a tenth of a percent

With Planck 2018 parameters ($H_0 = 67.36$, $\Omega_m = 0.3153$, $\Omega_b = 0.0493$, $\Omega_\gamma = 5.4 \times 10^{-5}$), integrating to $z = 10^6$ gives Table 1.

Table 1: The comoving sound horizon from the acoustic integral versus Planck 2018.

Quantity	This calculation	Planck 2018	Agreement
$r_s(z_*)$, last scattering ($z_* = 1089.80$)	144.2 Mpc	144.4 Mpc	0.1%
$r_s(z_{\text{drag}})$, drag epoch ($z_{\text{drag}} = 1059.94$)	146.9 Mpc	147.1 Mpc	0.2%

Proposition 4.1 (Sound-first horizon). *The comoving acoustic horizon at last scattering, computed solely from the subluminal sound speed (2) integrated against the pre-recombination expansion (3), reproduces the observed BAO ruler to sub-percent accuracy. The scale is fixed at z_* , the instant of photon decoupling.*

The registry standing wave $\delta_b \sim A j_0(kr_s)$ has its first node at $kr_s = \pi$, the $\ell = 0$ fundamental of the spherical-Bessel ladder (Paper I); with $r_s = 147.05$ Mpc, $k_1 = \pi/r_s = 0.0214 \text{ Mpc}^{-1}$ and the wiggle period $\Delta k = 2\pi/r_s = 0.0427 \text{ Mpc}^{-1}$.

5 Demarcation

Established physics: the photon–baryon sound speed (2); the acoustic horizon integral (3); the neutrino radiation contribution; recombination at $z_\star \approx 1090$ and drag at $z_{\text{drag}} \approx 1060$; the Planck horizon 144–147 Mpc; the $\sim 100 h^{-1}$ Mpc BAO feature.

Framework synthesis: the identification of the BAO ruler with the fundamental radial mode $j_0(kr_s)$ of the outermost breathing enclosure (ID7257/ID7259/ID7733), and the reading of $z_\star$ as the freezing instant supplying the acoustic clock for $H_0 = \dot{R}/R$ (ID10867).

Predictions: because the horizon is computed not fitted, any revision of Ω_b , Ω_m , or N_{eff} propagates deterministically into r_s ; a measured r_s drifting beyond the parameter uncertainties would wound the enclosure-acoustics identification at its most defensible node.

6 Limits of the construction

The integral uses fixed recombination epochs rather than a full Saha–Peebles ionization history, adequate at present precision. The result confirms the *scale* of the ruler, not the framework’s interpretation; mainstream cosmology already computes this horizon, and the added content is interpretive. Most importantly, sound-before-light does not remove the hot Big Bang: a sound horizon exists only because there was a hot, dense, ringing plasma, so the premise requires the hot early universe—the central honesty developed in Paper XII.

7 Conclusion: the stage is set, in sound

The ruler that anchors all cosmological distance measurement is a frozen sound wave whose size follows from first principles to a tenth of a percent. Read through the spine, it is the fundamental radial mode of the outermost breathing enclosure, frozen at decoupling: sound builds the ruler, light carries the snapshot. Paper XII places the breathing enclosure itself on this stage and asks what survives contact with the full CMB, BAO, and supernova distance data.

References

- [1] Trawin, D. A., *TZPID Canonical Equation Master Registry*, TZPIDProof archive (2026).
- [2] Trawin, D. A., *Nested Hyperspherical Enclosures*, TZPID Gold Spine Series, Paper I (2026).
- [3] Planck Collaboration, *Planck 2018 results. VI*, Astron. Astrophys. **641**, A6 (2020). arXiv:1807.06209.
- [4] Eisenstein, D. J., et al., *Detection of the baryon acoustic peak*, Astrophys. J. **633**, 560 (2005).
- [5] DESI Collaboration, *DESI DR2 Results II* (2025). arXiv:2503.14738.
- [6] Hu, W., and Sugiyama, N., *Small-scale cosmological perturbations*, Astrophys. J. **471**, 542 (1996).
- [7] Dodelson, S., *Modern Cosmology*, Academic Press (2003).